Engineering Electromagnetics II
Homework #4
48 points total

ECE 332 – Spring 2026

Due Friday, June 5, 2026

1. A 1-m long dipole is excited by a 1 MHz current with an amplitude of 12 A. What is the average power density radiated by the dipole at a distance of 5 km in a direction that is 45° from the dipole axis? (8 points)

2. The normalized radiation intensity of a certain antenna is given by

$$F(\theta) = \exp(-20\theta^2) \text{ for } 0 \le \theta \le \pi,$$

   where θ is in radians. Determine: (15 points total)

   - the half power beamwidth, (5 points)
   - the pattern solid angle, (5 points)
   - the antenna directivity. (5 points)

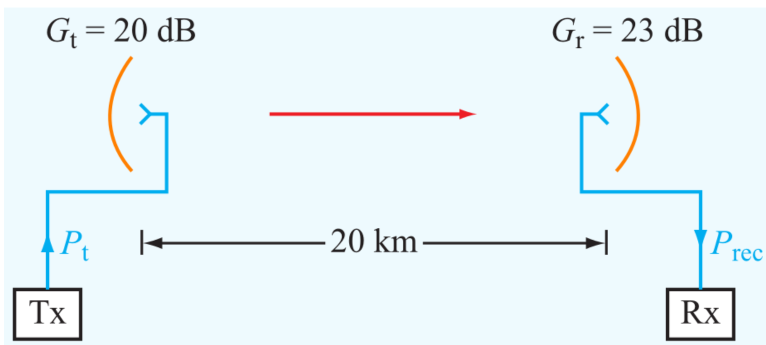
3. For a dipole antenna of length $l = \lambda/4$, (15 points total)

   - determine the directions of maximum radiation, (5 points)
   - obtain an expression for $S_{\max}$, (5 points)
   - generate a plot of the normalized radiation pattern $F(\theta)$. (5 points)

4. Determine the effective area of a half-wave dipole antenna at 100 MHz, and compare it with its physical cross section if the wire diameter is 2 cm. (5 points)

5. A half-wave dipole TV broadcast antenna transmits 1 kW at 50 MHz. What is the power received by a home television antenna with a 3 dB gain if located at a distance of 30 km? (5 points)

6. Consider the communication system with all components properly matched shown below.



Determine the following if $P_t = 10\,\text{W}$ and $f = 6\,\text{GHz}$.

(a) What is the power density at the receiving antenna (assuming proper alignment of antennas)?

(b) What is the received power?

(c) If $T_{\text{sys}} = 1000\,\text{K}$ and the receiver bandwidth is 20 MHz, what is the signal-to-noise ratio in decibels?

7. An antenna with a circular aperture has a circular beam with a beamwidth of 3° at 20 GHz.

(a) What is the antenna directivity in dB?

(b) If the antenna area is doubled, what will be the new directivity and new beamwidth?

(c) If the aperture is kept the same as in (a), but the frequency is doubled to 40 GHz, what will the directivity and beamwidth become then?

8. Find and plot the normalized array factor and determine the half-power beamwidth for a 5-element linear array excited with equal phase and a uniform amplitude distribution. The interelement spacing is $3\lambda/4$.